About This Course

ITDS271: Computer and Communication Security

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3rd Part
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Course Learning Outomes (CLO)

- CLO1: อธิบายหลักการและแนวคิดเกี่ยวกับ Cyber Security พร้อมทั้งมีความเข้าใจในหลักการเรื่องวิทยาการรหัสลับ ความปลอดภัยของแอปพลิเคชัน การรักษาความปลอดภัยฐานข้อมูล และการรักษาความปลอดภัยเครือข่าย
- CLO2: สามารถนำความรู้ที่เรียนไปใช้ในการระบุความเสี่ยง และแก้ปัญหาตัวอย่างด้าน Cyber Security ที่ได้รับมอบหมาย
- CLO3: สามารถประยุกต์ใช้หลักการของความมั่นคงปลอดภัยไซเบอร์ที่ได้เรียนมา เพื่อวิเคราะห์และแก้ปัญหาจากงานที่ได้รับมอบหมายและสามารถให้ข้อเสนอแนะได้อย่างเหมาะสม
- CLO4: มีความตระหนักถึงความรับผิดชอบทางจริยธรรมวิชาชีพทางด้าน Cyber Security

Course Plan and Topics

Mahidol University

Security Concepts and Cryptography

Week 1-5

- Security concpts and priciples
- Security awareness
- Basic cryptography:
  - Symmetric Encryption
  - Asymmetric Encryption
  - Hash Functions

Cybersecurity Applications

Week 6-15

- OS security
- Software security
- Web security
- Network security
- Email security
- Identification/Authentication
- Malware

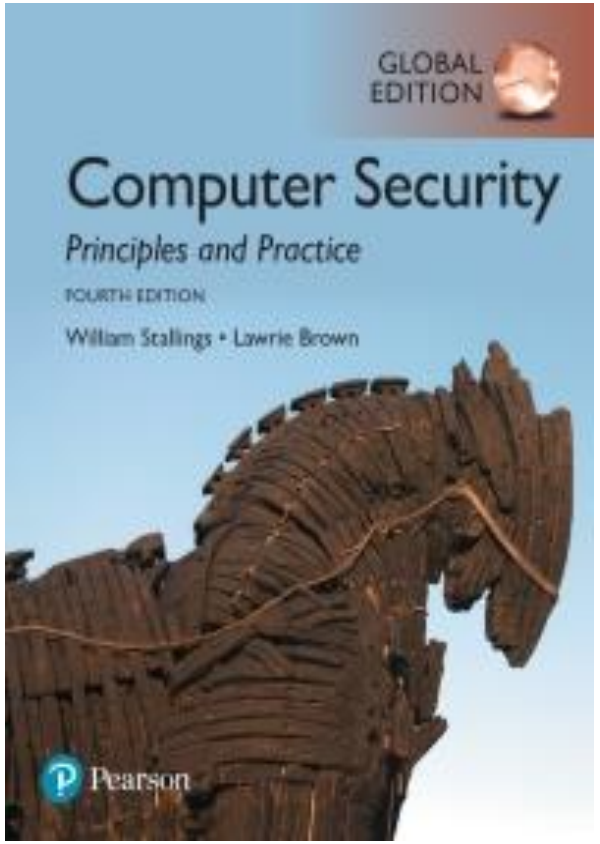
Course Evaluation

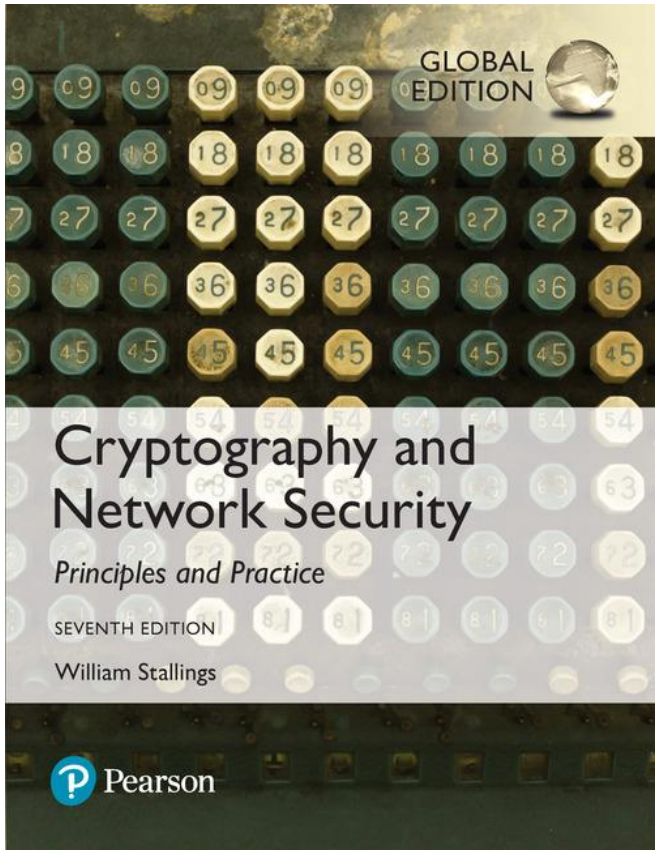
• Quiz (3)	15%
• Assignment	15%
• Midterm Exam	35%
• Final Exam	35%
Total:	**100%**

Course Outline and Textbooks

- Updated course outline will be posted on **MyCourses**
  - be sure to check it out from time to time

- **2 textbooks** are used and recommended for further study about the topics





Sick Leave and Business Leave

- ***Need to submit a leave form + evidence***
- **Sick leave** requires medical certificate (must be submitted within one week)
- **Business leave** considers only business related to the Faculty and university (must be submitted before the absent class)

Lecture 1: Security Overview

ITCS312: Computer and Communication Security

Dolvara Gunatilaka, PhD.

Today's Topics

- **Cybersecurity trends**
- Security definition and terminologies
- Computer security and the CIA triad
- Security attacks
- Security controls
- Social engineering

Cybersecurity Trends: Some Statistics …

- **Global Cost of Cybercrime:** projected to reach **$10.5 trillion annually by 2025**. [Source: Cybersecurity Ventures, BD Emerson, DeepStrike]

- **Frequency of Cyber Attacks:**
  - Cyberattacks occur at a rate of **over 2,200 times daily**, with someone falling victim approximately **every 39 seconds**. [Source: University of Maryland, Embroker]
  - Organizations experienced an average of 1,636 cyberattacks per week in Q2 2024, [Source: Check Point Research]

- **Data Breach:**
  - **83% of organizations** have experienced more than one data breach [Source: IBM Security]
  - Average time to identify and contain a data breach: **277 days**. [Source: SentinelOne]

This clear and accelerating danger, making cybersecurity vital for everyone!

Why Cybersecurity is More Critical Than Ever

- **Financial Impact:** Direct costs from data breaches, ransom payments, recovery efforts, and legal fees can be devastating
- **Reputational Damage:** Loss of customer trust, negative publicity, and harm to brand image can have long-lasting effects
- **Operational Disruption:** Attacks can halt business operations, supply chains, and critical services, leading to significant downtime and economic losses
- **Regulatory Compliance:** Stricter data protection laws (e.g., GDPR/PDPA) impose severe penalties for non-compliance and breaches
- **Personal Privacy:** Individuals face risks of identity theft, financial fraud, and privacy violations

Jobs in Cybersecurity

***Blue Team**: Defensive Security*

- Manage everyday security operation
- Detection, prevention, and response

- Cybersecurity engineer
- IT auditor
- SOC analyst (Tier 1, 2, 3)
- Incident responder
- Digital forensic analyst

***Red Team**: Offensive Security*

- Simulate real-world attacks
- Identify vulnerabilities

- Penetration tester / Ethical hacker

- Malware analyst
- Threat researcher

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Security Definition

- The protection of valuable items called **"assets"** from harm, theft, and unauthorized use

Security in Life

No one should be able to:

- Break into my house
- Harm me or my properties
- Steal my stuffs
- Use my properties without permissions
- ...

Security in CyberLife

No one should be able to:

- Break into my computer
- Attack my computer or damage my computer data
- Steal my information
- Use my resources without my permission
- ...

Security Terminologies

- Resources/Assets
- Vulnerability
- Threats
- Attacks
- Controls
- Risks

Terminologies: Assets and Vulnerability

- **Resources/Asstes:** what we are protecting
  - Can be tangible or non-tangible, e.g., hardware, software, data, infrastructure, people, etc.
- **Vulnerability:** weakness or flaws
  - Holes or gaps in security
  - Can exist in hardware, software, network configurations, human processes, or policies
  - E.g., coding errors, misconfigurations, weak password
  - 3 categories of vulnerability: the resources can be
    - Corrupted → loss of integrity
    - Leaky → loss of confidentiality
    - Unavailable → loss of availability

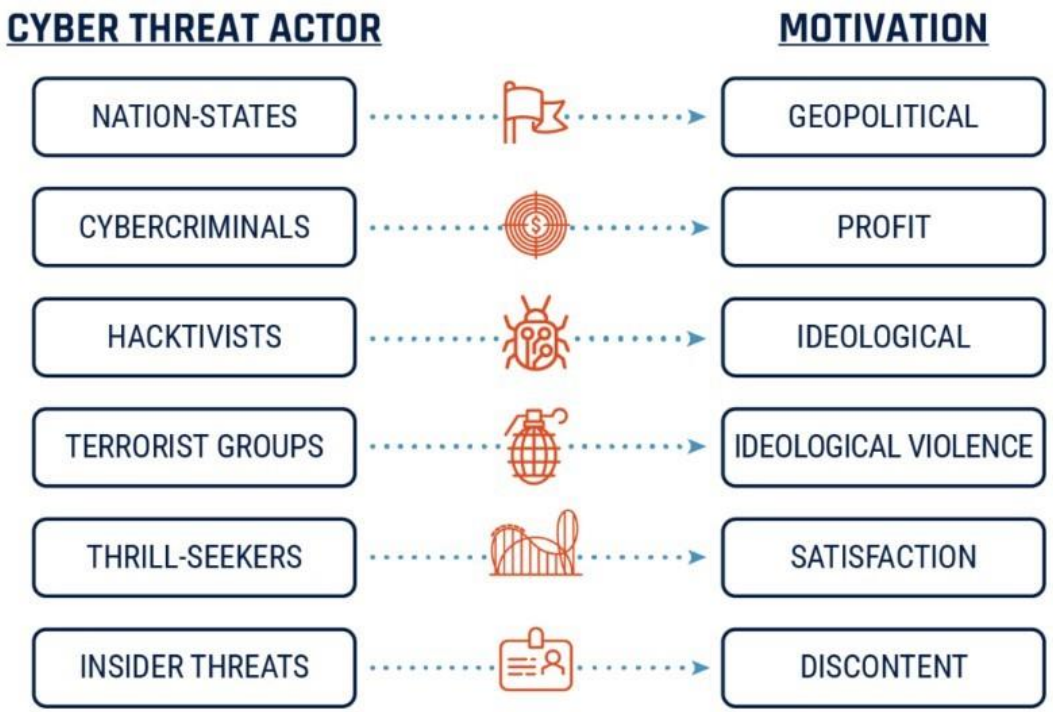
Terminologies: Threats

- **Threats:** harmful events that may cause loss/harm to the assets
  - Something that *could* exploit a **vulnerability**
  - E.g., a natural disaster that could disrupt data center, a new ransomware spreading across the internet
- Categories of threats:
  - **Human-Made Threats:**
    - **Accidental/Negligent:** Employee mistakes, misconfigurations, data entry errors
    - **Malicious:** Cybercriminals, hackers, nation-state actors, disgruntled employees (e.g., malware, phishing, denial-of-service attacks).
  - **Natural Threats:**
    - Floods, earthquakes, fires, power outages, severe weather.

Terminologies: Threat Agent

Threat Agent is an individual, a group, an organization that can initiate a threat, and want to exploit the assets of an organization

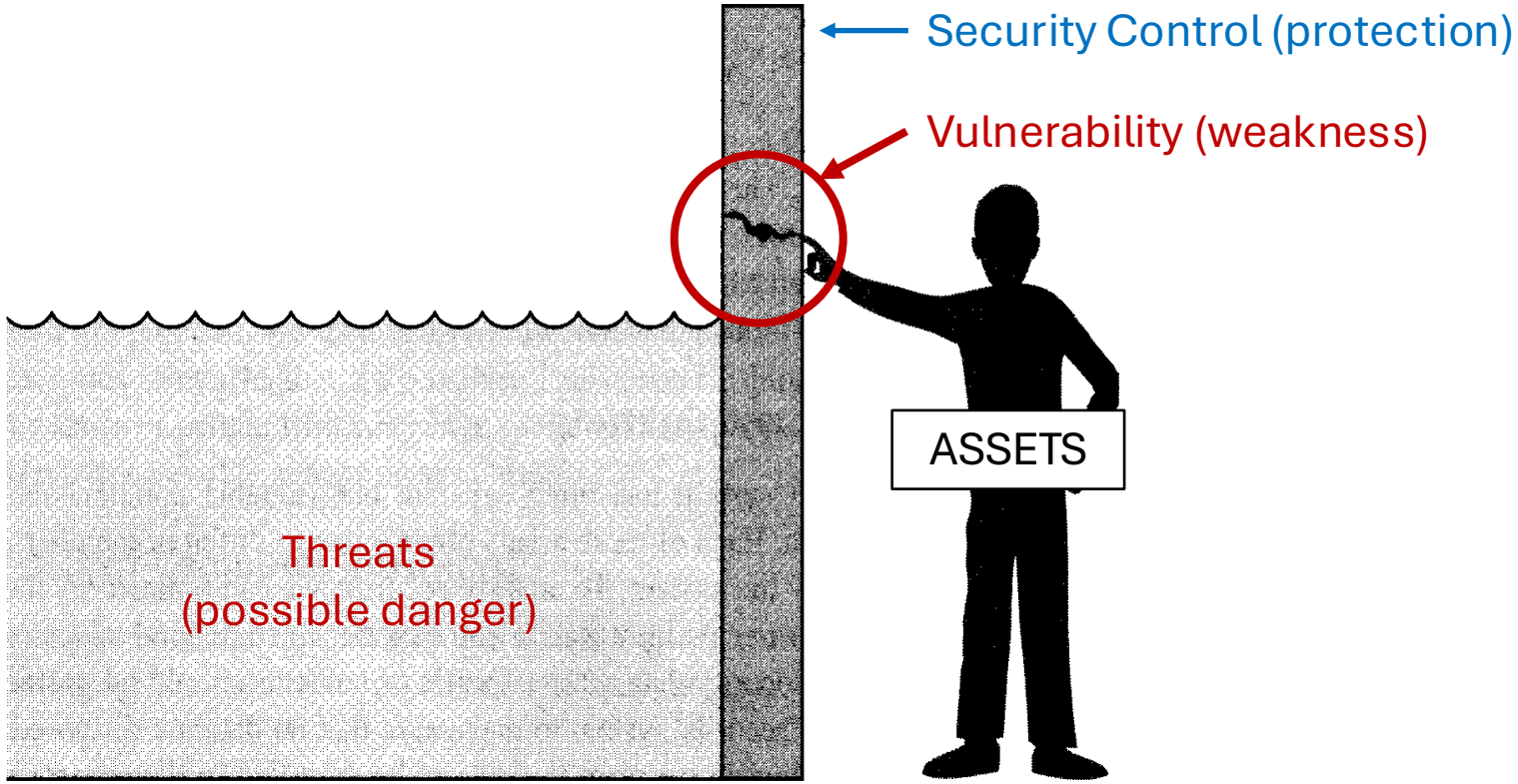
Potential **risk** posed by threat actor = ***capabilities + intention + past activities***



Terminologies: Attacks and Controls

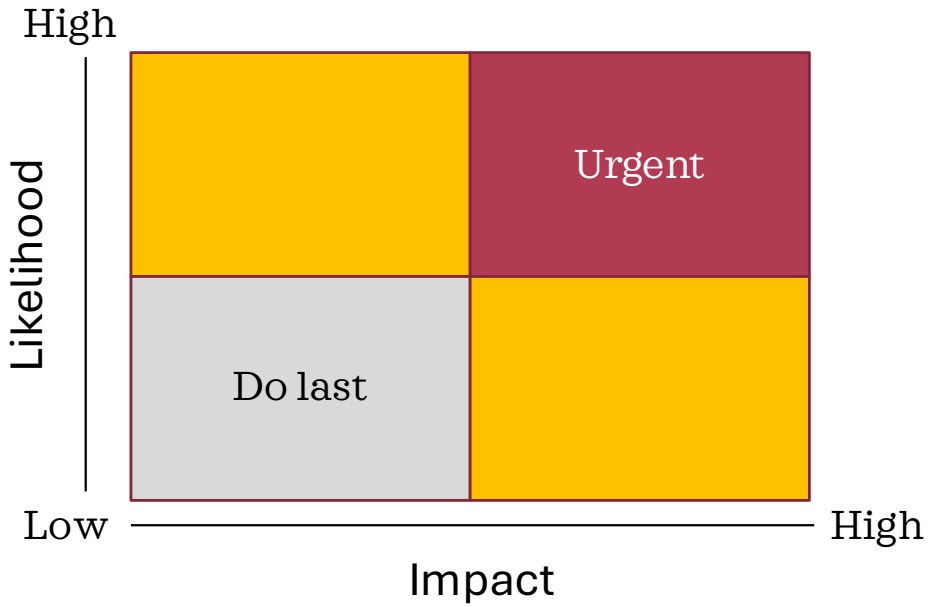
- **Attacks:** actual act of a **threat** exploiting a **vulnerability** to cause harm to an asset
  - Attack is a *deliberate* attempt to evade security services and violate security policy
  - Active adversary
  - E.g., malware attack, brute-force attack, DoS (Denial of Service)
- **Controls**: measures implmented to mitigate or remove the likelihood or impact of vulnerabilities, block threats
  - Can be actions, devices, procedures, or techniques
  - E.g., CCTV, firewall, data backups
- **Should we employ all and every possible security control to the system?**
  - No! That's not possible → **Why?**

Terminologies: Threats, Control, Vulnerabilities



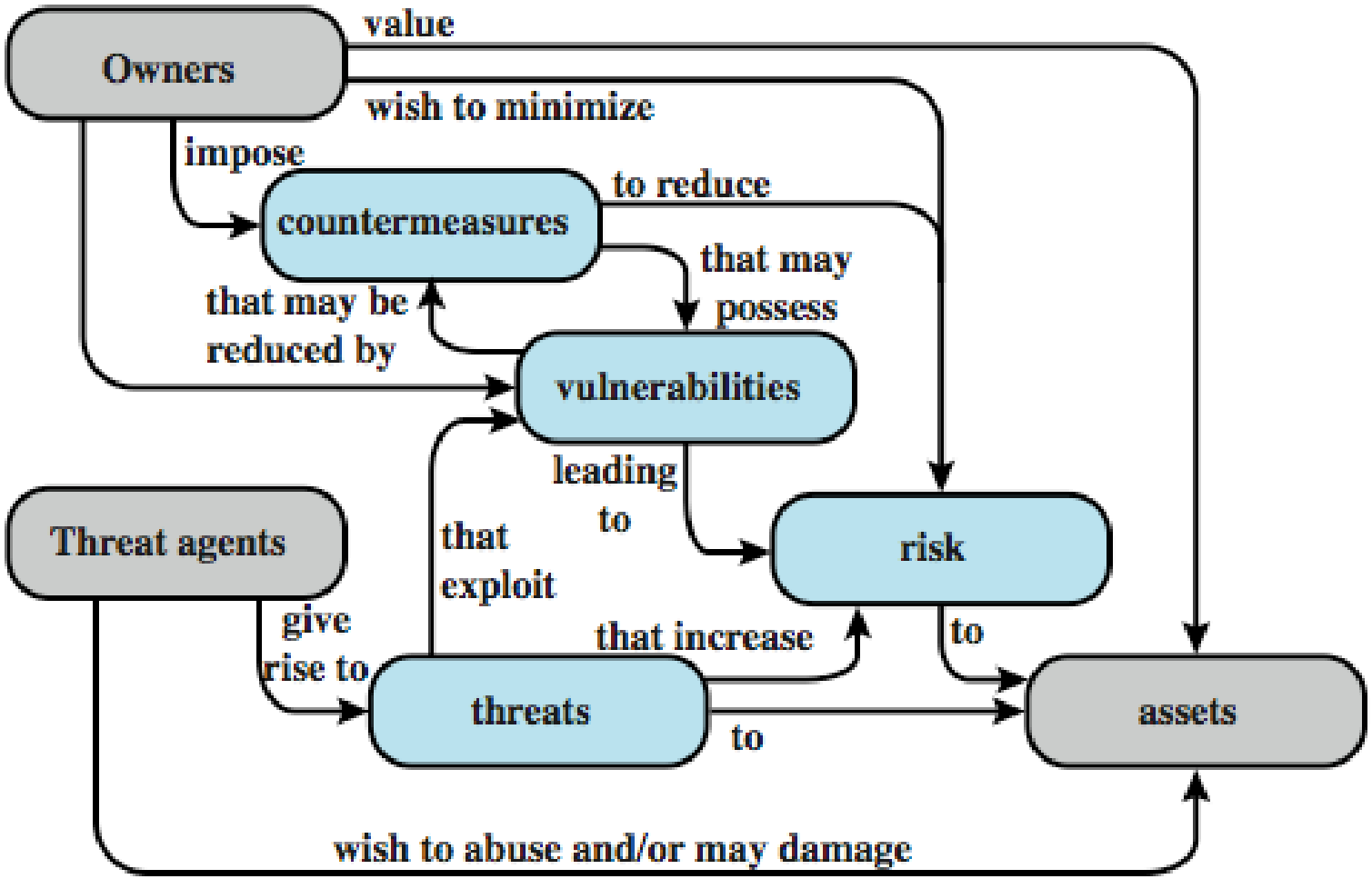
Terminologies: Risks

- **Risk:** The ***probability*** of loss and demage to an asset resulting from a <u>threat</u> *exploiting* <u>vulnerability</u>
  - Expressed as a combination of the ***likelihood*** of an event occurring and the ***impact*** of the event
- **Risk management:** the process to mitigate risks to acceptable level
  - Identify risks
  - Assess and prioritize risks
  - Treat risks
  - Monitor and review risks



Risk treatment:
- Avoid (Eliminate)
- Mitigate (Reduce)
- Transfer
- Accept

Security Concept and Relationship



Q&A

An outdated software system contains a buffer overflow flaw that can be exploited by malicious actors.

What is this an example of?

A. Threat

B. Control

C. Attack

D. Vulnerability

Today's Topics

- Cybersecurity trends
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- **Computer security and the CIA triad**
- Security attacks
- Security controls
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Compter Security / Cybersecurity

Computer Security

the protection afforded to an automated <u>information system</u> in order to attain the applicable objectives of preserving: Integrity, Availability, and Confidentiality

of information system resources; hardware, software, firmware, information/data, and telecommunications [NIST 800-12]

A more modern term: **Cybersecurity**

Preservation of confidentiality, integrity and availability of information in the Cyberspace [ISO/IEC 27032]

CIA Triad Overview

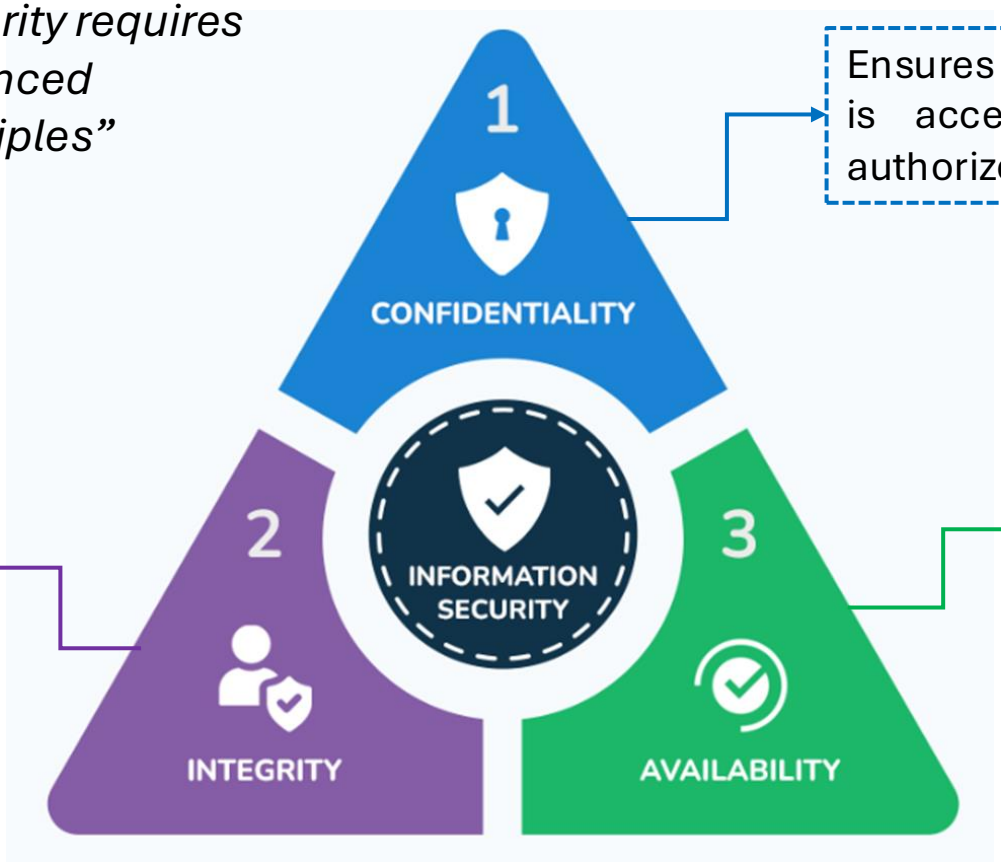
"Effective information security requires a comprehensive and balanced approach to all these principles"

C

Ensures that information is accessible only to authorized individuals.

I

Guarantees that information is accurate, complete, and has not been tampered with by unauthorized means



A

Ensures that authorized users can access the information and systems when needed.

Image: https://www.stationx.net/what-is-the-cia-triad/

CIA Triad: "C"

- Ensures that **data** is **accessible** <u>only</u> to **authorized** individuals
- Prevents sensitive data from being disclosed to unintended parties.
- ***Need-to-Know Principle*** **for Data Access:**
  - *"Only individuals whose job **requires** access to specific information should be granted access to that information"*
  - **Access Control:** Authentication and authorization

✅ **Examples of confidentiality measures:**

- Encryption
- Access control
- Physical security
- …

CIA Triad: "I"

- ***Data Integrity:*** ensures that information has not been **modified** or tampered in an unauthorized manner
  - Guarantees the trustworthiness and reliability of data
  - Key aspects:
    - Accuracy: data is correct and free from errors
    - Completeness: all necessary data is presented
- ***System Integrity:*** assure that a system work as intended and free from unauthorized manipulation

✅ **Examples of integrity measures:**

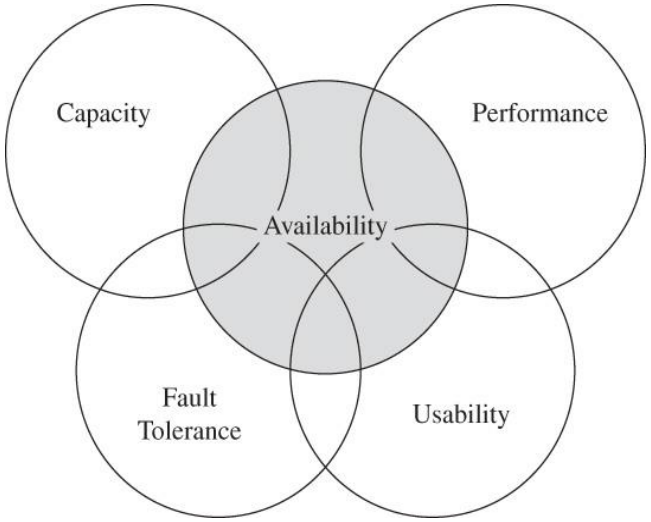
- Hashing
- Error correction/detection
- Version control
- …

CIA Triad: "A"

- Ensure that information and information systems are accessible and usable by authorized users when needed
- Services are <u>not</u> being **denied**
- Key aspects:
  - **System Uptime:** ensure servers, networks, and applications are functioning as they should and when they should
  - **Data Accessibility:** information can be retrieved and processed in a timely manner
  - **Resilience:** ability to withstand and recover from disruptions
  - **Fair Allocation:** ensure the system resources are distribute fairly → prevent starvation

✅ **Examples of availability measures:**

- Redundancy
- Backup
- Load balancing
- …



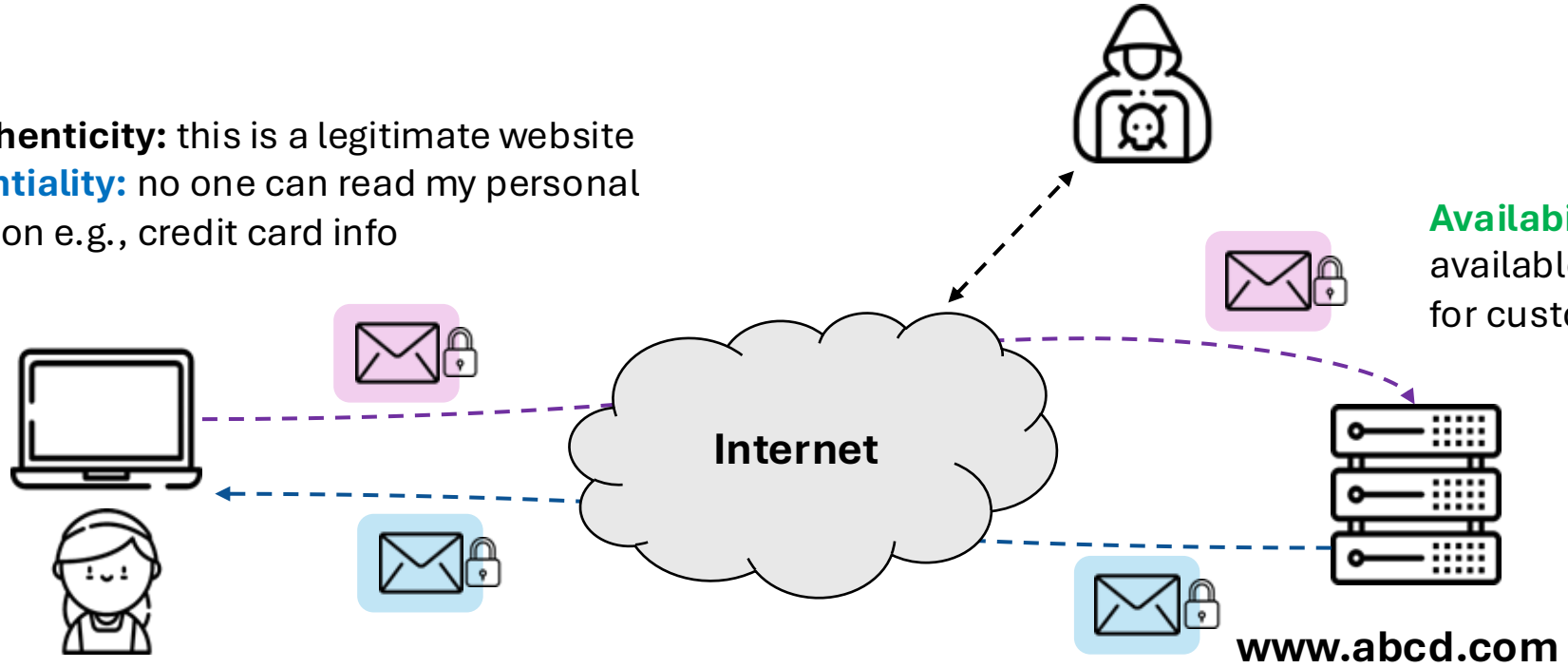
Beyond CIA: Authenticity and Accountability

- ***Authenticity***: ensure that the source of information or an action is genuinely who or what it claims to be
  - Identity **verification**: confirm legitimacy of an entity
  - **Trustworthiness**: establish confidence in the origin of data or actions

- ***Accountability:*** ensure actions within an information system to a specific individual or entity can be traced
  - **Traceability:** link actions to specific users or processes
  - **Non-Repudiation:** prevent user from denying that they performed a particular action
  - **Auditability:** ability to review logs and records to reconstruct events

CIA: Example Scenario

Peer authenticity: this is a legitimate website
Confidentiality: no one can read my personal information e.g., credit card info



Availability: the website is available to provide services for customers 24/7

Data origin authenticity: this message indeed comes from www.abcd.com
Integrity: no one has modified this message

Q&A

"An attacker sends a fake email pretending to be the CEO and instructs an employee to transfer funds to a fraudulent account."

What principle is being violated?

A. Confidentiality

B. Authenticity

C. Availability

D. Integrity

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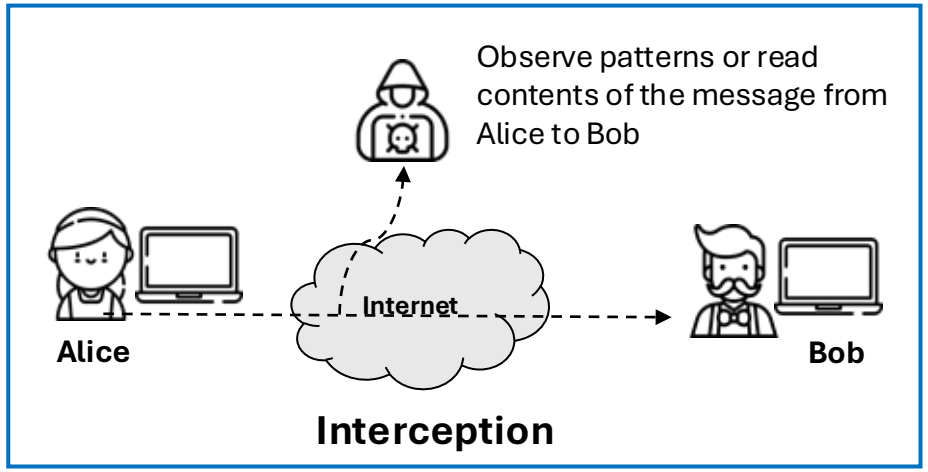
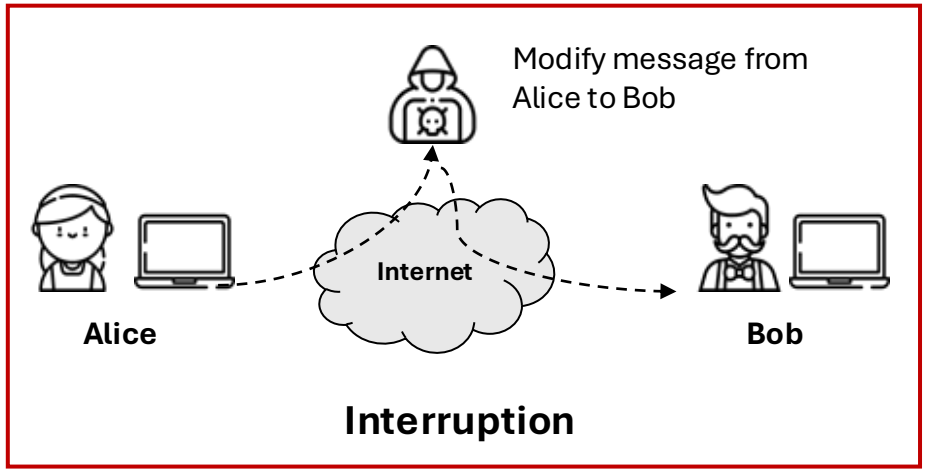
- **Security attacks**
- Security controls
- Social engineering

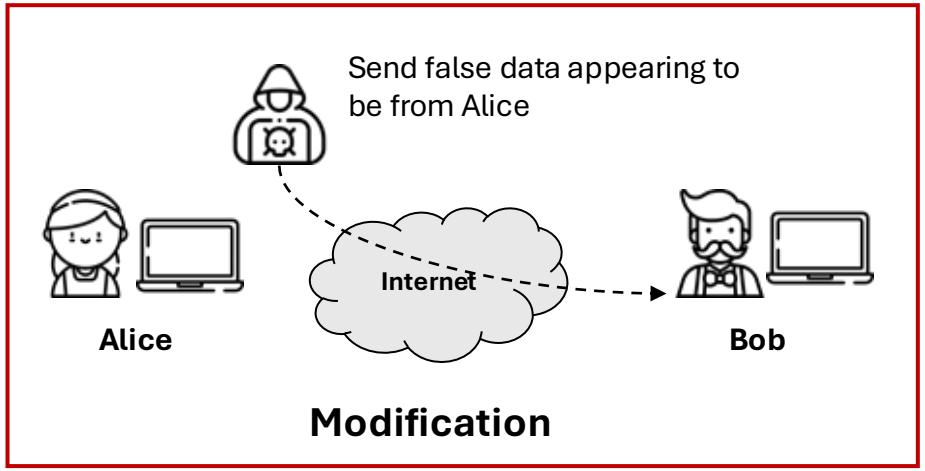
Security Attacks: Types

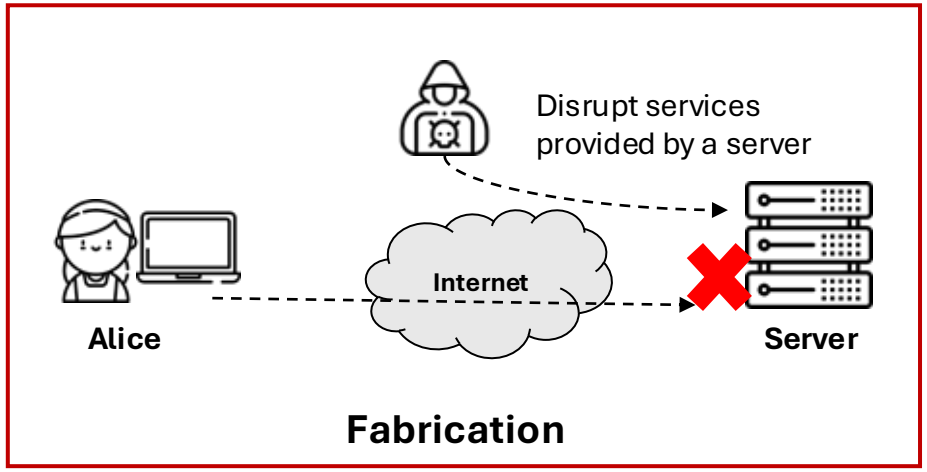
Passive Attack

Active Attack



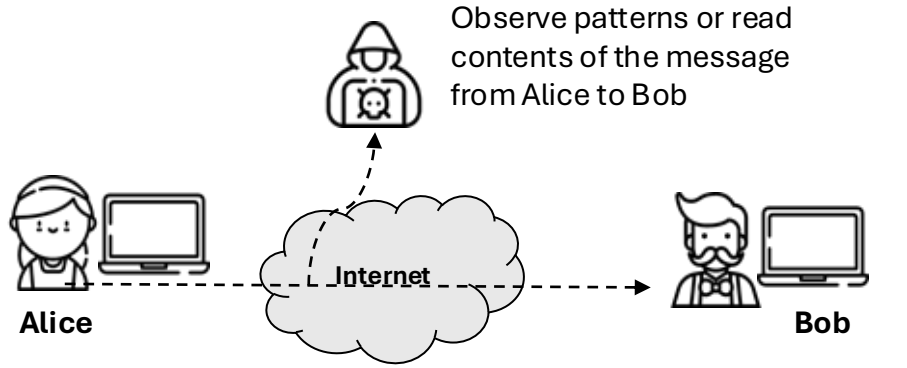


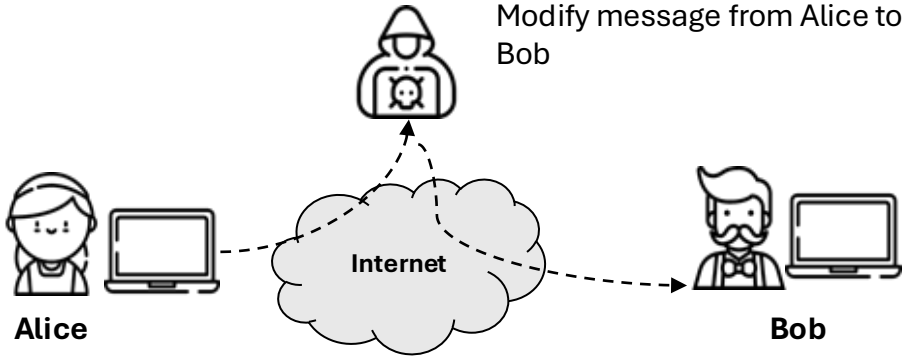




Security Attacks

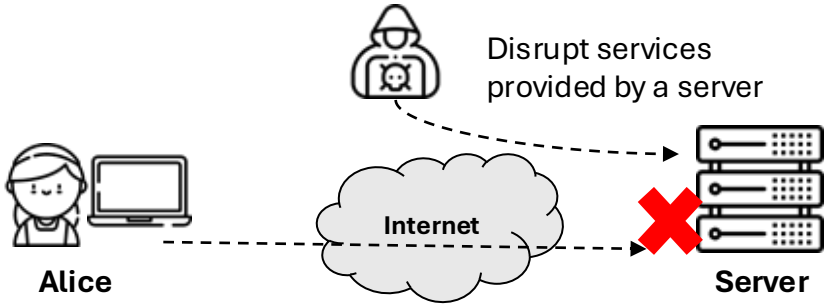
- **Interception:** attack on ***confidentiality***
  - Unauthorized access to resource or information
  - Eavesdropping: monitor communication to steal information, e.g., using packet sniffer
  - Illicit copying of files or program
- **Modification:** attack on ***integrity***
  - Resource is altered or tampered by unauthorized entity
  - Stop the flow of the message, modify it, and release it again



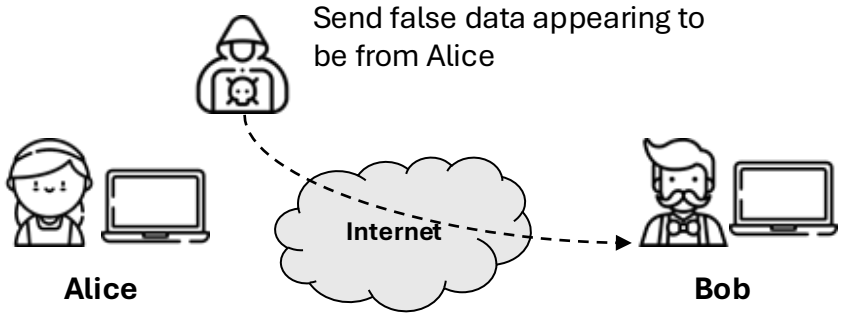


Security Attacks

- **Interruption:** attack on **availability** of system or data
  - Make the asset becomes <u>unavailable</u> or <u>unusable</u>
  - Destroy hardware (cutting fiber) or delete software
  - Corrupt packets in transit/jamming signals
  - ***Denial of service (DoS):*** overwhelm the server by using up its resources



- **Fabrication:** attack on **authenticity**
  - <u>False</u> data or activity are added or performed, and cannot be distinguished from real ones
  - Unauthorized assumption of other's identity
  - Generate and distribute objects under this identity



Attacker Types

- **Amateurs:** opportunistic attackers
  - Uses a password he/she found
- **Crackers** (evil doers) or Hackers
  - Enjoy simple challenge of trying to log in, just to see whether it can be done
- **Career criminals** (cybercriminal)
  - Understand the target of computer crime
  - Motivation is financial profit through illegal activities
- **Terrorists** (cyberterrorist)
  - State-supported spies and information warriors
  - Cause fear or significant disruption to critical infrastructure or public services

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Security Controls

Any measure put in place to reduce cybersecurity risks

Can be categorized based on their <u>functions</u>

- **Preventative Controls:** stop an attack before it happens
  - E.g., firewall, anti-malware, encryption, security awareness
- **Detective Controls:** identify an attack or security breach as it occurs or after it has occurred.
  - E.g., intrusion Detection Systems (IDS), CCTV, security audit
- **Corrective Controls:** restore systems or data to their original state after an incident and prevent recurrence.
  - E.g., data backups and recovery plans, disaster recovery plans, vulnerability remediation.

Security Controls

Security controls can also be categorized by how they are implemented

- **Physical control:** stop or block an attack by using something tangible
  - E.g., walls and fences, locks, human guards, sprinklers and other fire extinguishers
- **Technical control:** counter threats with technology (hardware or software)
  - Include passwords, OS, network protocols, firewalls, intrusion detection systems, encryption, network traffic flow regulators
- **Procedural or administrative control:** use a command or agreement that requires or advises people how to act
  - Laws, regulations, policies, procedures, guidelines, copyrights, patents, contracts, agreements

Security Controls

TYPE OF SECURITY CONTROLS	PREVENTATIVE	DETECTIVE	CORRECTIVE
PHYSICAL CONTROLS	> Fences > Gates > Locks	> CCTV > Surveillance Cameras	> Repair Physical damage > Reissue Access cards
TECHNICAL CONTROLS	> Firewalls > IPS > MFA > Antivirus	> IDS > Honeypots	> Vulnerability patching > Reboot a system > Quarantine a virus
ADMINISTRATIVE CONTROLS	> Hiring & termination policies > Separation of duties > Data classification	> Review access rights > Audit logs & unauthorized changes	> Implement a business continuity plan > Have an incident response plan

Source: https://www.infosectrain.com/blog/types-of-security-controls/

Q&A

Which of the following is an example of a *preventive physical* security control?

A. CCTV system

B. Firewall

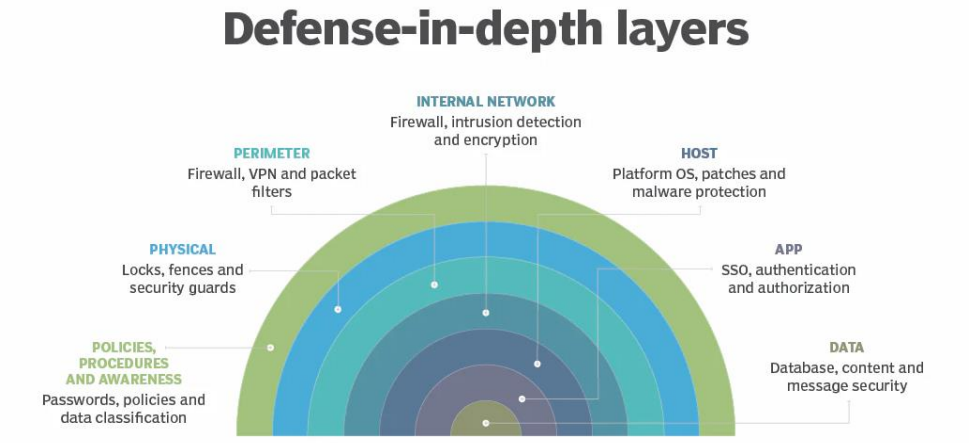
C. Security guard

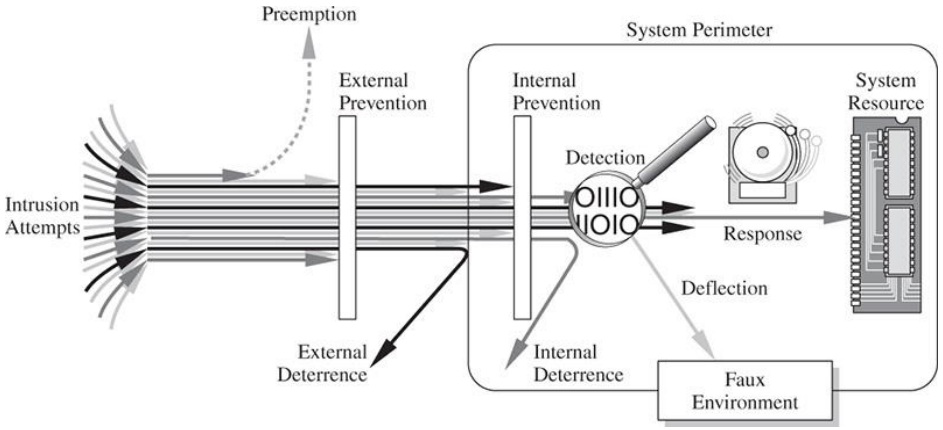
D. Alarm system

Security Controls: Layered Security

- No single control is foolproof
- Effective cybersecurity relies on a "***defense in depth***" strategy, which involves implementing multiple layers of security controls
- **Layered security**
  - If one control fails, another layer is in place to catch the threat
  - Increases the complexity and cost for attackers, making successful breaches less likely

Image: https://www.wiz.io/academy/defense-in-depth





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- **Social engineering**

Social Engineering

*"**Social engineering**" is the psychological manipulation of people into performing actions or divulging confidential information →* ***Hacking the human!***

Key principals

- Authority → CEO, CFO
- Intimidation → Penalty, “your will go to jail”
- Consensus → Bob, also do this
- Scarcity → limited offer, do it now
- Urgency → deadline
- Familiarity → trust

Phishing

- **Phishing**: A technique for attempting to acquire sensitive data, through a fraudulent solicitation in **email** or on a web site
  - Often sent to many recipients
  - *A digital form of <u>social engineering</u>*
- **Spearphishing**: <u>targeted</u> phishing attack aimed at specific individuals or organizations, using personalized information to increase credibility.
- **Whailing: a** type of spear phishing attack specifically targeting <u>high-profile individuals</u> like CEOs or executives.

Phishing

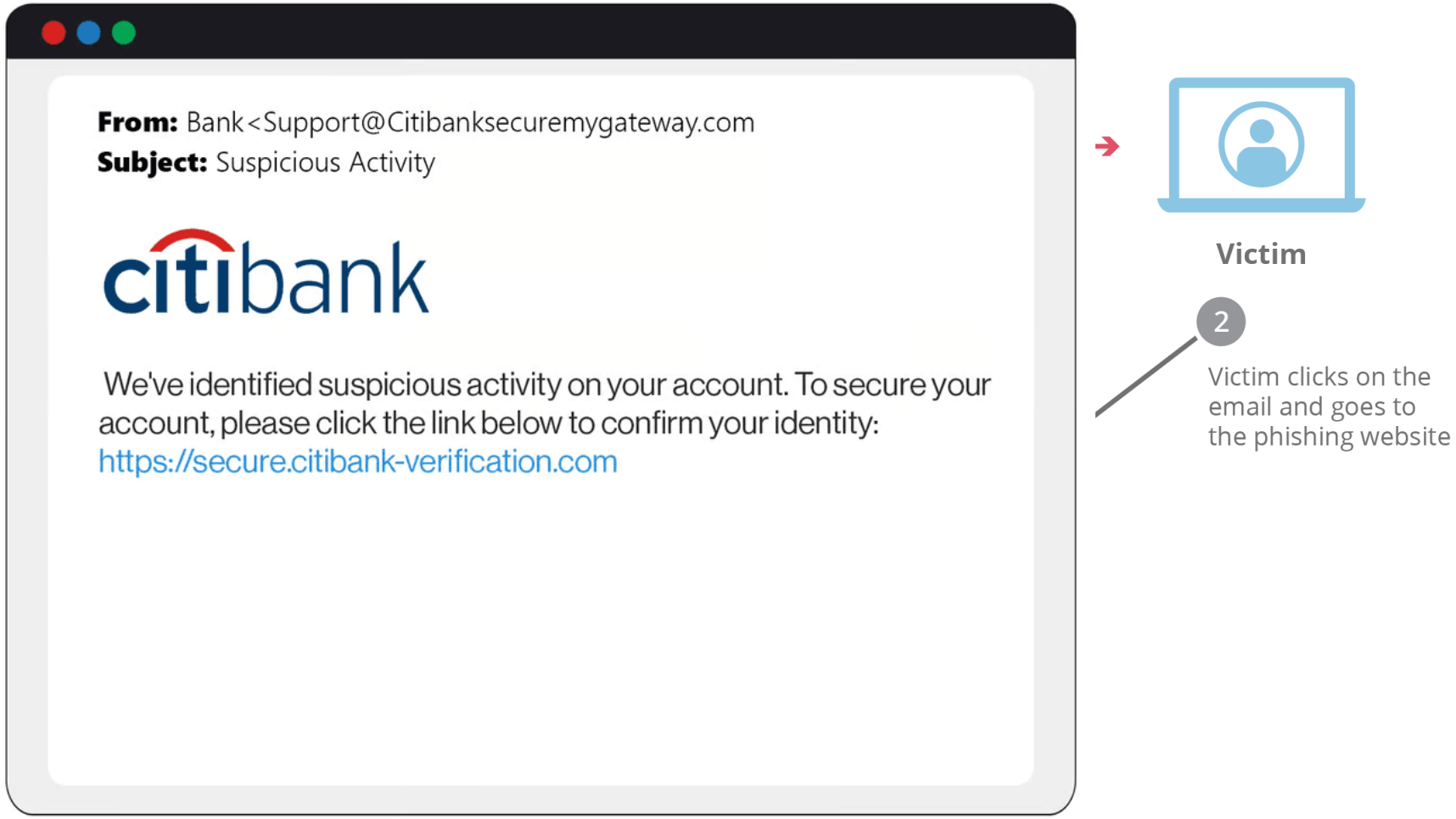


Image: https://www.cloudflare.com/learning/access-management/phishing-attack/

Types of Social Engineering

- **Vishing**: involve fraudulent phone calls or voicemails to obtain sensitive data

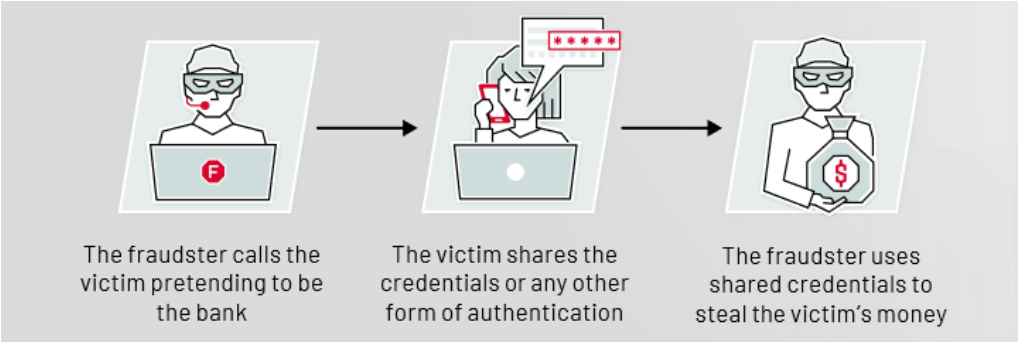
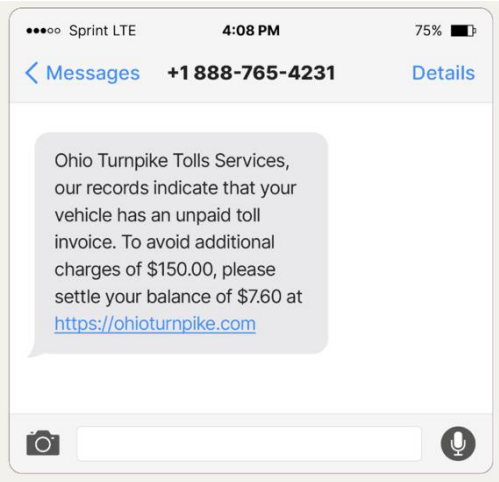


Image: https://www.threatmark.com/on-vishing-attacks-and-how-to-prevent-them/

- **Smishing**: use deceptive text messages (SMS) to trick individuals into revealing personal information or clicking malicious links.



Summary

- Cybersecurity trends: cybersecurity is more critical than ever!
- Security terminologies: asset, vulnerability, threat, attack, control, risk
- CIA triad: **C**onfidentiality, **I**ntegrity, **A**vailability
- Security attacks: Interception, modification, interruption, fabrication
- Security controls: no single control is foolproof. Effective cybersecurity relies on a "defense in depth" strategy
- Social engineering: digital form of social engineering → phishing

มหาวิทยาลัยมหิดล
Mahidol University

ห้ามมิให้นักศึกษาทำซ้ำ ดัดแปลง หรือใช้ประโยชน์จากงานอันมีลิขสิทธิ์ปรากฏอยู่ในระบบการศึกษาอิเล็กทรอนิกส์ (E-Learning) ของมหาวิทยาลัย ไม่ว่าจะทั้งหมดหรือบางส่วนโดยไม่ได้รับอนุญาตจากมหาวิทยาลัย นอกเหนือจากการศึกษาส่วนบุคคล ทั้งนี้ การทำซ้ำ ดัดแปลง หรือเผยแพร่ต่อสาธารณชนซึ่งงานอันมีลิขสิทธิ์ **จะมีโทษปรับตั้งแต่ 20,000 บาท ถึง 200,000 บาท และหากเป็นการกระทำเพื่อการค้า จะมีโทษจำคุกตั้งแต่ 6 เดือน ถึง 10 ปี หรือปรับตั้งแต่ 100,000 บาท ถึง 800,000 บาท หรือทั้งจำทั้งปรับ**

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